EXPERIMENT 06

```
clc;
clear;
close all;

% Simulation-Based Investigation of Microwave Semiconductor Devices
% and Frequency-Conversion Characteristics of a Microwave Mixer

%% 1. Gunn Diode Model

V = linspace(0,10,500);       % Voltage (V)

% Illustrative Gunn diode I-V model
I_Gunn = 0.02*V + 0.15*V.^2 .* exp(-V/1.8);

% Add a negative differential resistance region
I_Gunn = I_Gunn - 0.22*exp(-((V-4.5)/0.9).^2);
I_Gunn = I_Gunn - min(I_Gunn) + 0.02;

% Differential conductance
dIdV = gradient(I_Gunn,V);

%% 2. Schottky Diode Model

Vd = linspace(-0.5,0.5,500);
Is = 1e-8;                  % Saturation current (A)
n = 1.2;                    % Ideality factor
VT = 0.02585;               % Thermal voltage (V)

I_Schottky = Is*(exp(Vd/(n*VT))-1);
I_Schottky = max(min(I_Schottky,1e-2),-1e-2);
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%% 3. PIN Diode Frequency Response

freq = logspace(6,10,500);    % 1 MHz to 10 GHz
fGHz = freq/1e9;

% Illustrative normalized PIN diode response
fc_PIN = 1e9;
H_PIN = 1./sqrt(1+(freq/fc_PIN).^2);
H_PIN_dB = 20*log10(H_PIN);

%% 4. Microwave Mixer Frequency Conversion

% RF and local oscillator frequencies
fRF = 10e9;                % RF frequency = 10 GHz
fLO = 9.8e9;               % LO frequency = 9.8 GHz
fIF = abs(fRF-fLO);        % Intermediate frequency = 200 MHz
fSum = fRF+fLO;            % Sum frequency = 19.8 GHz

% Frequency spectrum representation
freq_components = [fIF fRF fLO fSum]/1e9;
amplitudes = [0.8 1.0 1.0 0.35];

%% 5. Mixer Time-Domain Signals

Fs = 100e9;                % Sampling frequency
T = 50e-9;                 % Simulation duration
t = 0:1/Fs:T-1/Fs;

RF = cos(2*pi*fRF*t);
LO = cos(2*pi*fLO*t);

% Idealized mixer output: product of RF and LO
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mixer_output = RF .* LO;

%% 6. FFT Spectrum of Mixer Output

N = length(mixer_output);
Y = abs(fft(mixer_output))/N;
Y = Y(1:floor(N/2)+1);
Y(2:end-1) = 2*Y(2:end-1);

fFFT = (0:floor(N/2))*Fs/N;

% Display spectrum up to 22 GHz
valid = fFFT <= 22e9;

%% 7. Conversion Loss versus LO Power

P_LO_dBm = linspace(-10,15,200);

% Illustrative conversion-loss model
P_opt = 7;                  % Assumed optimum LO power (dBm)
CL = 7 + 0.035*(P_LO_dBm-P_opt).^2;

% Assumed conversion-loss improvement near optimum LO drive
CL = max(CL,0);

%% 8. Create One Large Figure with Six Graphs

figure('Color','w', ...
    'Name','Microwave Semiconductor Devices and Mixer', ...
    'Units','normalized', ...
    'OuterPosition',[0 0 1 1]);
```

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tiledlayout(2,3, ...
    'TileSpacing','compact', ...
    'Padding','compact');

%% GRAPH 1: Gunn Diode I-V Characteristics

nexttile;
plot(V,I_Gunn,'b','LineWidth',2);
grid on;
xlabel('Voltage (V)');
ylabel('Current (A)');
title('1. Gunn Diode I-V Characteristics');

%% GRAPH 2: Gunn Diode Differential Conductance

nexttile;
plot(V,dIdV,'r','LineWidth',2);
hold on;
yline(0,'k--','Zero conductance');
grid on;
xlabel('Voltage (V)');
ylabel('dI/dV (A/V)');
title('2. Gunn Diode Differential Conductance');

%% GRAPH 3: Schottky Diode I-V Characteristics

nexttile;
plot(Vd,I_Schottky*1e3,'m','LineWidth',2);
grid on;
xlabel('Diode Voltage (V)');
ylabel('Current (mA)');
title('3. Schottky Diode I-V Characteristics');
```

```
%% GRAPH 4: PIN Diode Frequency Response

nexttile;
semilogx(fGHz,H_PIN_dB,'g','LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Normalized Response (dB)');
title('4. PIN Diode Frequency Response');

%% GRAPH 5: Mixer Frequency Components

nexttile;
stem(freq_components,amplitudes,'filled','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Relative Amplitude');
title('5. Microwave Mixer Frequency Components');
legend('IF, RF, LO and Sum Frequency', ...
    'Location','best');

%% GRAPH 6: Conversion Loss versus LO Power

nexttile;
plot(P_LO_dBm,CL,'k','LineWidth',2);
hold on;
[~,idxOpt] = min(CL);
plot(P_LO_dBm(idxOpt),CL(idxOpt),'ro', ...
    'MarkerFaceColor','r','MarkerSize',7);
grid on;
xlabel('LO Input Power (dBm)');
ylabel('Conversion Loss (dB)');
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```
title('6. Conversion Loss versus LO Power');
legend('Illustrative Model','Assumed Optimum', ...
    'Location','best');

%% 9. Overall Figure Formatting

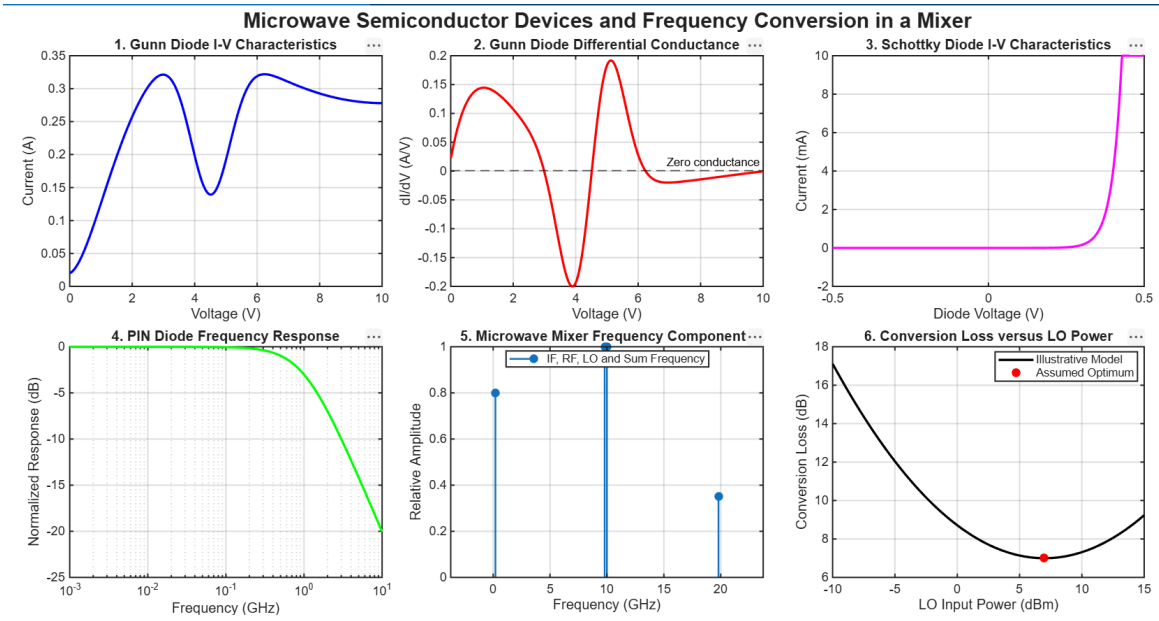
sgtitle(['Microwave Semiconductor Devices and ' ...
    'Frequency Conversion in a Mixer'], ...
    'FontSize',18,'FontWeight','bold');

ax = findall(gcf,'Type','axes');
set(ax,'FontSize',11,'LineWidth',1.2);

%% 10. Display Calculated Mixer Frequencies

fprintf('\n--- Microwave Mixer Results ---\n');
fprintf('RF Frequency       = %.2f GHz\n',fRF/1e9);
fprintf('LO Frequency       = %.2f GHz\n',fLO/1e9);
fprintf('Difference (IF)    = %.2f GHz\n',fIF/1e9);
fprintf('Sum Frequency      = %.2f GHz\n',fSum/1e9);
fprintf('Assumed LO optimum = %.2f dBm\n',P_LO_dBm(idxOpt));
fprintf('Minimum modeled conversion loss = %.2f dB\n',CL(idxOpt));

disp('Simulation completed: six graphs in one figure.');
```


Microwave Semiconductor Devices and Frequency Conversion in a Mixer
1. Gunn Diode I-V Characteristics
Current (A)
Voltage (V)
2. Gunn Diode Differential Conductance
dI/dV (A/V)
Zero conductance
Voltage (V)
3. Schottky Diode I-V Characteristics
Current (mA)
Diode Voltage (V)
4. PIN Diode Frequency Response
Normalized Response (dB)
Frequency (GHz)
5. Microwave Mixer Frequency Component
IF, RF, LO and Sum Frequency
Relative Amplitude
Frequency (GHz)
6. Conversion Loss versus LO Power
Illustrative Model
Assumed Optimum
Conversion Loss (dB)
LO Input Power (dBm)